Chapter 11. Relational Inference Matched to the Data Structure

Graphical models, Lecture 4 (hidden confounders, FCI and PAGs, nonlinear CI tests, PCMCI+, feedback)

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1 PC Assumes All Common Causes Are Observed

Premises used by PC: relations among the observed variables are represented by one DAG; no common cause of the analyzed variables is missing; the Markov condition and faithfulness hold and conditional independences are tested properly.

Problem: an unobserved common cause. Suppose $E \leftarrow U \rightarrow P$ with U a hidden lifestyle factor. The observed data contain only exercise E and blood pressure P. E and P are related, but they are not directly connected. Because PC assumes all common causes are observed, it may represent this relation as a direct edge. **If hidden common causes are possible, an edge found by PC cannot be interpreted as a direct relation.**

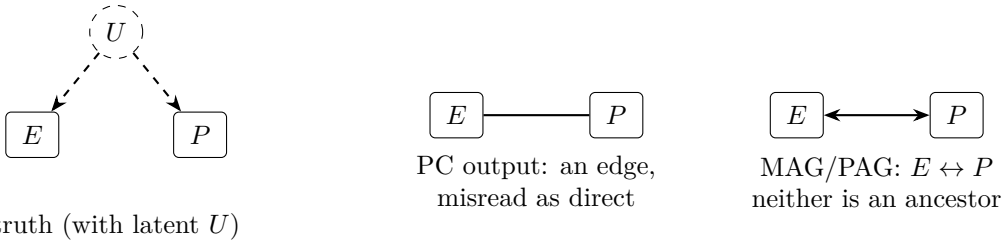


Figure 1: A hidden common cause. Marginalizing U leaves a dependence between E and P with no direct effect. A DAG over the observed variables must pick a direction; the ancestral-graph language of FCI has a bidirected edge for exactly this situation (identifiable when enough other variables are observed).

1.1 Observational data may not distinguish three explanations

Suppose only X and Y are observed and they are related.

Possible structure	Explanation	Observed conditional independences
$X \rightarrow Y$	X causes Y	none
$X \leftarrow Y$	Y causes X	none
$X \leftarrow U \rightarrow Y$	hidden common cause U	none

Additional observed variables and their conditional independences can exclude some explanations, but a single DAG is not always determined.

2 FCI Infers a PAG Instead of a DAG

2.1 Ancestral graphs and PAGs

When latent variables are marginalized out of a DAG, the conditional independences among the observed variables are represented by a **maximal ancestral graph** (MAG) with directed ($\rightarrow$) and bidirected ($\leftrightarrow$) edges: $X \rightarrow Y$ means X is an ancestor of Y; $X \leftrightarrow Y$ means neither is an ancestor of the other (a latent confounder). Markov-equivalent MAGs are summarized by a **partial ancestral graph** (PAG), in which an endpoint mark is shown only if it is common to all MAGs of the class; otherwise a circle $\circ$ is drawn.

	Object represented	Meaning of relations
PC's CPDAG	DAG equivalence class under the assumption of no missing common causes among analyzed variables	only directions common to all equivalent DAGs are fixed
FCI's PAG	equivalence class of observed-variable structures allowing hidden common causes	only endpoints common to all compatible structures are fixed

FCI (Fast Causal Inference) uses conditional independences but does *not* assume causal sufficiency. The circles and arrowheads of a PAG are not an incomplete result: they display the range that observational data can distinguish.

2.2 Reading PAG endpoints

PAG edge	How to read it
$X \to Y$	apart from selection bias, in every compatible structure X is an ancestor of Y and Y is not an ancestor of X
$X \leftrightarrow Y$	neither is an ancestor of the other; compatible with a structure with a hidden common cause
$X \circ\!\!\to Y$	it is certain that Y is not an ancestor of X; the mark at X is undetermined
$X \circ\!\!-\!\!\circ Y$	the two variables are adjacent but both endpoints are undetermined

An arrowhead means that variable is *not an ancestor* of the other. A circle may be a tail or an arrowhead depending on the compatible structure. $X \to Y$ in a PAG is an ancestral relation, not the single direct effect of a latent DAG.

2.3 FCI can still find the arrowheads of colliders

Suppose CI tests give the skeleton $X - Y - Z$ with $X \not\sim Z$, and $X \perp\!\!\!\perp Z$ unconditionally but $X \not\perp\!\!\!\perp Z \mid Y$. Without conditioning on Y, the path is blocked at the collider and X, Z are independent; conditioning on the collider opens the path and they become related. Hence FCI orients the arrowheads into Y:

$$X \circ\!\!\to Y \leftarrow\!\!\circ Z.$$

Whether X (or Z) is a cause of Y or shares a hidden cause with it remains open.

2.4 FCI checks wider separating sets, then orients conservatively

Algorithm: FCI

Edge removal.

1. Start from the complete graph on the observed variables.
2. If $X \perp\!\!\!\perp Y \mid S$ is found, remove $X - Y$ (as in PC's skeleton phase).
3. Widen the candidate separating sets beyond current neighbors to the variables on possible paths (Possible-D-Sep) and check again: because of hidden variables, a separating set may not lie among the current neighbors.

Endpoint determination.

1. Record the separating set used when an edge was removed.
2. Mark as colliders the middle variables not contained in the separating set.

3. Propagate only those endpoints that are forced, using the colliders and the PAG orientation rules (Zhang's rules R1–R10).

Undetermined endpoints remain circles $\circ$.

Remark: why neighbors are not enough

In a DAG, a non-adjacent pair is always separated by the parents of one of them, which are neighbors. In a MAG this fails: a separating set may have to contain a node W that reaches X only through a path whose inner nodes are all colliders (e.g. $W \leftrightarrow V \leftrightarrow X$ created by latent variables), and such a W need not be adjacent to X or Y. The set D-Sep(X, Y) collects such nodes; since it depends on the unknown MAG, FCI uses the computable superset Possible-D-Sep from the intermediate graph. PC, which only searches among neighbors, would wrongly keep the edge. The extra tests are the price of not assuming causal sufficiency.

Like PC, FCI starts from conditional independences, but checks the additional separation possibilities needed because of hidden variables. **RFCI** (really fast CI) skips most of the Possible-D-Sep search and runs targeted extra tests on triples, reducing computation at the cost of a slightly less informative (but still sound) output.

3 When Basic Assumptions Break, Methods Change

Basic assumption	Situation where it fails	Candidate methods
all common causes observed	unobserved common causes exist	FCI, RFCI
the chosen test captures the relation	partial correlation misses nonlinear relations	KCI, distance correlation
independent observations	autocorrelation and time lags	PCMCI+
no cycles in the graph	feedback among variables	CCD, cyclic LiNGAM

The choice of algorithm depends less on the data size than on **which basic assumptions can be maintained**.

3.1 Nonlinear relations: change the independence test

Partial correlation measures linear relations.

Example: dependence with zero correlation

Let $X \sim \text{Unif}(-1, 1)$ and $Y = X^2 + \varepsilon$ with $\varepsilon \perp\!\!\!\perp X$. Then $\text{Cov}(X, Y) = \mathbb{E}[X^3] - \mathbb{E}[X]\mathbb{E}[X^2] = 0$ because the distribution of X is symmetric, although Y depends strongly on X. A Fisher-z test would declare them independent and PC would remove the edge.

CI test	When to use
Fisher-z	linear conditional relations in continuous Gaussian data
kernel CI test (KCI)	general nonlinear conditional relations (based on cross-covariance operators in reproducing-kernel spaces)
distance correlation	nonlinear relations without strong assumptions on the distribution; $\text{dCor}(X, Y) = 0$ iff $X \perp\!\!\!\perp Y$

The search procedures of PC and FCI stay the same; only the CI test is replaced. Nonparametric tests are more flexible but lose power with small samples or many conditioning variables, and are computationally heavier.

3.2 Time series: PCMCI+ for lagged relations

Ordinary PC treats rows as independent, but in a time series the past of a variable is related to its present. Relations of interest are

$$X^j_{t-\tau} \longrightarrow X^k_t, \qquad \tau = 0, 1, \ldots, \tau_{\max}.$$

Algorithm: PCMCI+ (outline)

1. Among the past values of all variables, first reduce the candidate parents that may be related to each present value (a PC-type condition-selection step on lagged links).
2. Re-test each candidate link conditioning on the other past parents of both endpoints (momentary conditional independence, MCI), which removes autocorrelation effects.
3. Distinguish time-lagged relations from same-time (contemporaneous) relations and represent them in a time-series graph; contemporaneous links are oriented with PC-type rules.

Example output: $X^1_{t-2} \to X^2_t$ and $X^2_{t-1} \to X^3_t$. Feeding time series into ordinary PC can create false edges through autocorrelation: two independent but strongly autocorrelated series often show large sample correlations. Check the maximal lag $\tau_{\max}$ and stationarity.

3.3 Feedback: the DAG assumption fails

Economic and biological systems can have feedback such as $X_1 \to X_2 \to X_3 \to X_1$.

Method	Core idea
CCD	uses conditional independences to find relations common to the equivalence class of cyclic graphs
cyclic LiNGAM	uses linearity and non-Gaussian errors to further identify the cyclic structure

A linear cyclic SEM $X = BX + \epsilon$ is read as an equilibrium of a dynamic system, requiring $I - B$ invertible (stability typically needs the spectral radius of B below 1). CCD allows cycles, but directions the observational data cannot distinguish remain undetermined. Before using a cyclic algorithm, decide whether the feedback is *simultaneous* or actually a *time-lagged* relation that PCMCI+ should handle.

4 Choosing a Method for the Data Structure

Feature of the data	Method	Main points to check
hidden common causes suspected	FCI or RFCI	read PAG circles and arrowheads separately
nonlinear relations expected	PC/FCI with nonparametric tests	test power and computing time
time lags and autocorrelation	PCMCI+	maximal lag and stationarity
simultaneous feedback possible	CCD or cyclic LiNGAM	distinguish cycles from time lags

Summary

- With hidden confounders, PC's edges are not direct relations; FCI outputs a PAG whose arrowheads mean "not an ancestor" and whose circles mean "undetermined".
- FCI widens the search for separating sets (Possible-D-Sep) because separating sets may lie outside the neighbors; RFCI trades some information for speed.
- Nonlinear relations need kernel or distance-based CI tests; time series need lag-aware methods (PCMCI+); feedback needs cyclic methods (CCD, cyclic LiNGAM).

5 Selected exercises

Motivation. The exercises show that a hidden common cause can reproduce a direct effect exactly and how one extra observed variable separates the two, quantify the spurious correlation between independent autocorrelated series, and justify the equilibrium reading of a cyclic SEM.

Exercise 11.1 (A hidden common cause mimics a direct effect)

Let $U \sim N(0,1)$ be latent, $X = aU + \epsilon_X$, $Y = bU + \epsilon_Y$, with $U, \epsilon_X, \epsilon_Y$ independent, $\epsilon_X \sim N(0, \sigma_X^2)$, $\epsilon_Y \sim N(0, \sigma_Y^2)$, $\sigma_Y^2 > 0$.
(a) Find c and $\tau^2 > 0$ such that the SEM $X' \sim N(0, a^2 + \sigma_X^2)$, $Y' = cX' + \eta$, $\eta \sim N(0, \tau^2)$ independent of X', has the same distribution as (X, Y).
(b) Add an observed $Z \sim N(0,1)$, independent of $(U, \epsilon_X, \epsilon_Y)$, with $X = dZ + aU + \epsilon_X$, $d \neq 0$. Show that $Z \perp\!\!\!\perp Y$, whereas in any Gaussian SEM $Z \to X \to Y$ with nonzero coefficients Z and Y are dependent.

Solution [Supplement].

(a)

Step 1. (X, Y) is bivariate Gaussian with mean 0, $\mathrm{Var}X = a^2 + \sigma_X^2$, $\mathrm{Var}Y = b^2 + \sigma_Y^2$ and $\mathrm{Cov}(X, Y) = ab$. A centered Gaussian law is determined by its covariance matrix.

Step 2. In the second SEM, $\mathrm{Cov}(X', Y') = c(a^2 + \sigma_X^2)$ and $\mathrm{Var}Y' = c^2(a^2 + \sigma_X^2) + \tau^2$. Matching gives

$$c = \frac{ab}{a^2 + \sigma_X^2}, \qquad \tau^2 = b^2 + \sigma_Y^2 - \frac{a^2 b^2}{a^2 + \sigma_X^2} = \sigma_Y^2 + \frac{b^2 \sigma_X^2}{a^2 + \sigma_X^2} > 0.$$

So no procedure using (X, Y) alone can tell $X \leftarrow U \to Y$ from $X \to Y$ (or, symmetrically, from $Y \to X$).

(b)

Step 3. $Y = bU + \epsilon_Y$ is a function of (U, ϵ_Y), which is independent of Z; hence $Z \perp\!\!\!\perp Y$.

Step 4. If $X = dZ + \epsilon'_X$ and $Y = cX + \eta$ with independent errors and $c, d \neq 0$, then $\mathrm{Cov}(Z, Y) = cd \neq 0$, so Z and Y are dependent.

The extra variable turns the question into a conditional-independence pattern: $Z \perp\!\!\!\perp Y$ together with $Z \not\perp\!\!\!\perp X$ and $X \not\perp\!\!\!\perp Y$ is exactly the evidence FCI uses to orient $Z \circ\!\!\to X \leftarrow\!\!\circ Y$; the arrowhead at X says that X is not an ancestor of Y, which rules out $X \to Y$. ■

Exercise 11.2 (Independent but autocorrelated series)

Let $X_t = \phi X_{t-1} + u_t$ and $Y_t = \phi Y_{t-1} + v_t$ be stationary AR(1) processes, $|\phi| < 1$, where (u_t) and (v_t) are i.i.d. $(0, 1)$ sequences independent of each other. Let $C_n = \frac{1}{n}\sum_{t=1}^{n} X_t Y_t$. Show that $\mathbb{E}C_n = 0$ and

$$n\mathrm{Var}(C_n) \longrightarrow \gamma(0)^2 \frac{1+\phi^2}{1-\phi^2}, \qquad \gamma(0) = \frac{1}{1-\phi^2},$$

whereas for $\phi = 0$ the limit is $\gamma(0)^2 = 1$.

Solution [Supplement].

Step 1. The stationary solution $X_t = \sum_{m\geq 0} \phi^m u_{t-m}$ has mean 0 and autocovariance $\gamma(h) = \phi^{|h|}/(1-\phi^2)$; the same holds for Y. The processes are independent, so $\mathbb{E}(X_t Y_t) = \mathbb{E}X_t\,\mathbb{E}Y_t = 0$.

Step 2. By independence of the two processes,

$$\mathrm{Cov}(X_s Y_s, X_t Y_t) = \mathbb{E}(X_s X_t)\,\mathbb{E}(Y_s Y_t) = \gamma(t-s)^2.$$

Step 3. Hence

$$n\mathrm{Var}(C_n) = \frac{1}{n}\sum_{s=1}^{n}\sum_{t=1}^{n} \gamma(t-s)^2 = \sum_{|h|<n} \Big(1 - \frac{|h|}{n}\Big)\gamma(h)^2.$$

Since $\sum_h \gamma(h)^2 < \infty$, dominated convergence gives the limit

$$\sum_{h\in\mathbb{Z}} \gamma(h)^2 = \gamma(0)^2\Big(1 + 2\sum_{h\geq 1}\phi^{2h}\Big) = \gamma(0)^2\Big(1 + \frac{2\phi^2}{1-\phi^2}\Big) = \gamma(0)^2\,\frac{1+\phi^2}{1-\phi^2}.$$

The sample variances converge to $\gamma(0)$, so the sample correlation $r_n \approx C_n/\gamma(0)$ has variance about $\frac{1}{n}\cdot\frac{1+\phi^2}{1-\phi^2}$ instead of $\frac{1}{n}$; for $\phi = 0.9$ this is 19 times larger. A Fisher-z test that treats the rows as independent therefore declares many spurious edges, which PCMCI+ avoids by conditioning on the past. ■

Exercise 11.3 (Equilibrium of a linear cyclic SEM)

Let B be a $p \times p$ matrix with spectral radius $\varrho(B) < 1$.
(a) Show that $I - B$ is invertible and $(I-B)^{-1} = \sum_{m\geq 0} B^m$.
(b) Show that for a fixed ϵ and any start $X^{(0)}$, the iteration $X^{(t+1)} = BX^{(t)} + \epsilon$ converges to $(I-B)^{-1}\epsilon$.
(c) For the feedback loop $X_1 = bX_2 + \epsilon_1$, $X_2 = aX_1 + \epsilon_2$, show that $\varrho(B) < 1$ iff $|ab| < 1$, and find the equilibrium.

Solution [Supplement].

(a)

Step 1. By Gelfand's formula $\varrho(B) = \lim_m \|B^m\|^{1/m}$ for any matrix norm, so for $\varrho(B) < q < 1$ there is m_0 with $\|B^m\| \leq q^m$ for $m \geq m_0$. Hence $\sum_m \|B^m\| < \infty$ and $\sum_m B^m$ converges.

Step 2. $(I-B)\sum_{m=0}^{M} B^m = I - B^{M+1} \to I$, so $(I-B)\sum_{m\geq 0} B^m = I$; the same holds with the factors in the other order.

(b) By induction, $X^{(t)} = B^t X^{(0)} + \sum_{m=0}^{t-1} B^m \epsilon$. The first term tends to 0 by Step 1 and the second to $(I-B)^{-1}\epsilon$.

(c) Here $B = \begin{pmatrix} 0 & b \\ a & 0 \end{pmatrix}$ has characteristic polynomial $\lambda^2 - ab$, so its eigenvalues have modulus $\sqrt{|ab|}$ and $\varrho(B) < 1$ iff $|ab| < 1$. Solving the two equations,

$$X_1 = \frac{\epsilon_1 + b\epsilon_2}{1-ab}, \qquad X_2 = \frac{a\epsilon_1 + \epsilon_2}{1-ab}.$$

The observed data are the fixed point of a dynamic system in which each shock is passed around the loop repeatedly ($\sum_m B^m$); if $|ab| \geq 1$ the shocks are amplified and no stable equilibrium exists. ■